

Multiple regression

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The game

Multiple regression

Interpreting MLR

Measuring Predictive Ability

Significance in MLR

Interpreting MLR II

The logo for the game show "Deal or No Deal" is centered. It features the words "DEAL" and "NO DEAL" in a bold, black, sans-serif font, separated by a small white square containing the word "or". The text is set against a yellow rectangular background with a metallic, brushed-metal texture. The entire logo is set against a dark background with a grid of glowing orange and red squares, creating a sense of depth and movement. A bright, glowing orange light source is visible behind the logo, casting a warm glow and creating a lens flare effect. Two thin, curved white lines sweep across the background, adding a dynamic feel to the composition.

DEAL or NO DEAL



26 briefcases, with hidden amounts in each case



Dollar values range from \$0.01 to \$1 million





Contestant chooses one case



In each “round” they must open a certain number of the remaining cases



6 CASES TO OPEN

Using the information about what cases have been opened...



The “banker” makes an offer to the contestant to buy their case



The contestant decides whether to make a “deal” and accept the offer or not

- If they accept (“deal”), they walk away with the money the banker offered.
- If they decline (“no deal”), they have to open more cases in the next round.
- If they have opened all of the remaining cases, they go home with whatever was in the briefcase they chose at the beginning of the game.



DEAL OR NO DEAL

So how does the banker come up with their offer?



Data set

- 77 rounds from 9 games of Deal or No Deal

Let's use the average of the remaining cases to predict the banker's offer:

```
model <- lm(offer ~ average, data=deal)
summary(model)
```

Call:

```
lm(formula = offer ~ average, data = deal)
```

Residuals:

Min	1Q	Median	3Q	Max
-74426	-32524	13614	28068	94948

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.152e+04	8.409e+03	-2.56	0.0125 *
average	6.854e-01	5.141e-02	13.33	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 41880 on 75 degrees of freedom

Multiple R-squared: 0.7032, Adjusted R-squared: 0.6993

F-statistic: 177.7 on 1 and 75 DF, p-value: < 2.2e-16

What **else** might help predict the banker's offer?



Two simple regression models

Predictor	Equation	R^2
average	$\widehat{\text{offer}} = -21524.65 + .69 \cdot \text{average}$	0.70
round	$\widehat{\text{offer}} = 4440 + 13766 \cdot \text{round}$	0.21



Two simple regression models

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Can we put these predictor variables together to make a better prediction than either of them individually?

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Using multiple predictors to predict the offer

The simple regression models were:

$$Y_i = \beta_0 + \beta_1(\text{round}) + \epsilon_i$$

and

$$Y_i = \beta_0 + \beta_1(\text{average}) + \epsilon_i.$$

The multiple regression model is

$$Y_i = \beta_0 + \beta_1(\text{round}) + \beta_2(\text{average}) + \epsilon_i.$$

```
model <- lm(offer ~ average + round, data=deal)
summary(model)
```

Call:

```
lm(formula = offer ~ average + round, data = deal)
```

Residuals:

Min	1Q	Median	3Q	Max
-48143	-19511	-1611	19314	58726

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-8.438e+04	7.294e+03	-11.57	<2e-16 ***
average	6.780e-01	3.049e-02	22.24	<2e-16 ***
round	1.325e+04	1.123e+03	11.81	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 24830 on 74 degrees of freedom

Multiple R-squared: 0.8971, Adjusted R-squared: 0.8943

F-statistic: 322.5 on 2 and 74 DF, p-value: < 2.2e-16

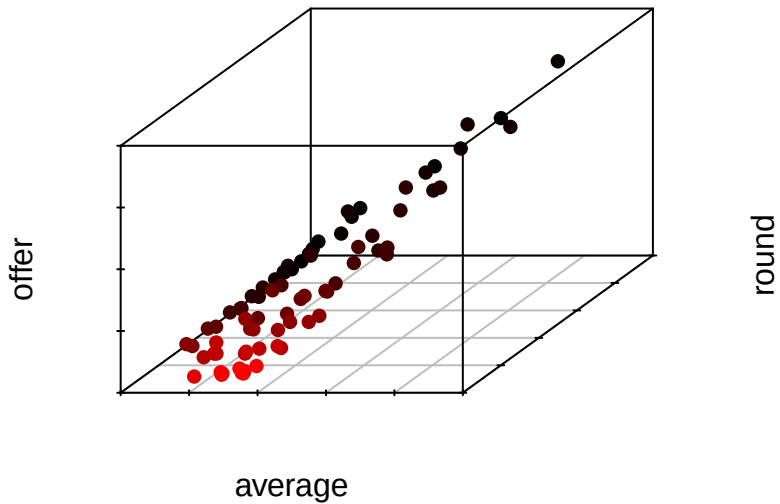
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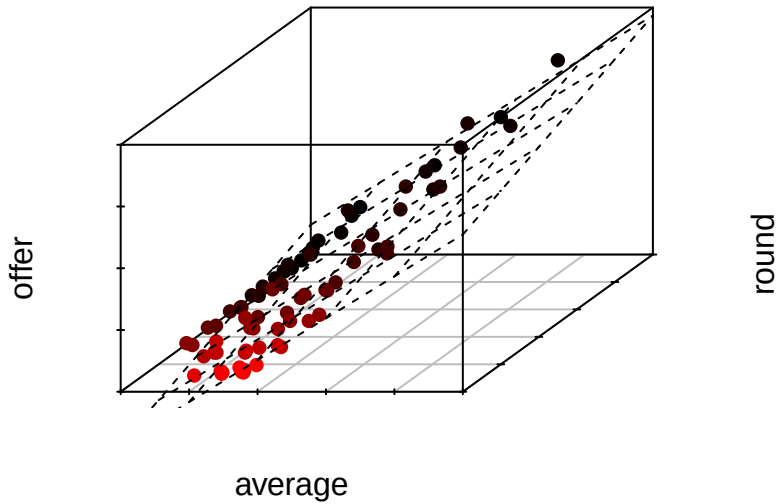
$$\widehat{\text{offer}} = -84382.34 + 0.68(\text{average}) + 13252.72(\text{round})$$

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$$\widehat{\text{offer}} = -84382.34 + 0.68(\text{average}) + 13252.72(\text{round})$$

We can use this to make predictions like we would for a simple regression!





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Interpreting the coefficients: intercept

Let's interpret the intercept coefficient of -84382.34 :

- The predicted banker's offer when average on the board is 0 and the round number is 0 is -84382.34 .



Interpreting the coefficients: intercept

Let's interpret the intercept coefficient of -84382.34 :

- The predicted banker's offer when average on the board is 0 and the round number is 0 is -84382.34 .
- This is not a meaningful number on its own in this case, since neither variable can ever be 0 in the game. (The intercept might be interpretable for other models.)



Interpreting the coefficients: average

Let's interpret the average of 0.68:

- Among players that are at the same point in the game, each dollar in the average of the amounts left on the board increases our predicted banker's offer by \$0.68.



Interpreting the coefficients: average

Let's interpret the average of 0.68:

- Among players that are at the same point in the game, each dollar in the average of the amounts left on the board increases our predicted banker's offer by \$0.68.
- If we compared two players that are at the same point in the game but differ by \$1 in the average amount left on the board, the player with the higher average would be predicted to have a banker's offer that is \$0.68 higher.



Interpreting the coefficients: round

Let's interpret the round coefficient of 13252.72:

- Among players that have the same average amount left on the board, the banker's offer is predicted to increase by \$13252.72 in every round.



Interpreting the coefficients: round

Let's interpret the round coefficient of 13252.72:

- Among players that have the same average amount left on the board, the banker's offer is predicted to increase by \$13252.72 in every round.
- If we compared two players that have the same average amount left on the board but one player is one round further in the game than the other, the player who is further along would be predicted to have a banker's offer that is \$13252.72 higher.



What's the difference?!

- “The predicted effect of a \$1 increase in the average amount left on the board” and “the predicted effect of a \$1 increase in the average amount left on the board, holding the round number constant” really are **two different things**.



What's the difference?!

- “The predicted effect of a \$1 increase in the average amount left on the board” and “the predicted effect of a \$1 increase in the average amount left on the board, holding the round number constant” really are **two different things**.
- The relationship between X_1 and Y may change when we **control for** (i.e., add to the model) another predictor X_2 .



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R^2 for multiple regression models

Predictors	R^2
average	0.70
round	0.21
average and round together	0.90

- R^2 indicates the proportion of the variance of the banker's offers that is explained by the average amount left on the board and what round the player is in, together.



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- R^2 will **always increase** when you add more variables to the model, but the R^2 values don't add.
- $\sqrt{R^2} = R = 0.95$ is the **correlation between $\hat{Y}$ and Y** (the correlation between our predictions of the banker's offer and the banker's actual offers).



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- R reports two R^2 – multiple and adjusted
- Multiple R^2 is easier to interpret (previous slide)
- Adjusted R^2 is similar, but tries to better estimate the true population R^2 by accounting for the number of parameters in the model
- Unless specified otherwise, R^2 will refer to multiple R^2



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- Residuals will always average out to 0
- If the residuals are **Normally distributed with equal variance** approximately 95% of the residuals will be less than $\pm 2s_e$.
- In our combined model, $s_e = 24829.3$
- Adding good predictors will reduce s_e – less uncertainty about future values

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The overall null hypothesis for a regression model

The following are equivalent ways to express the overall null hypothesis with k predictor variables:

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- $\text{cor}(\hat{Y}, Y) = 0$ (in the population)
- $\beta_1 = \beta_2 = \dots = \beta_k = 0$ (i.e., all coefficients are 0 except the intercept)
- There is no linear relationship between Y and *any* of the X 's
- Predictions from this model are no better than predicting $\bar{Y}$ for every case – $s_e = s_y$, the sample SD of Y

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Conducted using an F -test.

In this model, the overall p -value is very small, so we reject the overall null hypothesis and conclude that yes, we have statistical significance and that this model does have some predictive power.

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summary(model)
```

```
Call:
```

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lm(formula = offer ~ average + round, data = deal)
```

```
Residuals:
```

Min	1Q	Median	3Q	Max
-48143	-19511	-1611	19314	58726

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-84382.3382	7293.8401	-11.6	<2e-16 ***
average	0.6780	0.0305	22.2	<2e-16 ***
round	13252.7167	1122.5356	11.8	<2e-16 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 24800 on 74 degrees of freedom
```

```
Multiple R-squared:  0.897, Adjusted R-squared:  0.894
```

```
F-statistic: 322 on 2 and 74 DF, p-value: <2e-16
```

The p-value for the overall hypothesis test is in the very last row (beginning with "F-statistic")



Testing individual coefficients

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- X_i has to bring some *new* predictive ability to the model
- The regression output calculates the p -value for us for testing the null hypotheses $\beta_i = 0$.
- If we reject this null hypothesis for a coefficient, we say that X_i is a statistically significant predictor of Y in the model.



Testing individual coefficients

Be careful interpreting this test!

- We can only interpret multiple regression coefficients *in context*, based on the other variables in the model.



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- Remember: Statistical significance is not practical significance. Smaller p-value doesn't mean the added variable is a stronger predictor!



Testing individual coefficients

Be careful interpreting this test!

- We can only interpret multiple regression coefficients *in context*, based on the other variables in the model.
- Remember: Statistical significance is not practical significance. Smaller p-value doesn't mean the added variable is a stronger predictor!
- Finally, make sure the question is relevant. Sometimes we include a variable in the model to make sure we are adjusting for confounders or “lurking variables”, not just to improve predictions.

Deciding whether a new variable is worth it



- If the p -value for a coefficient is significant (generally, < 0.05) then we can be confident that adding this new variable does help in predicting Y , and the new variable is **statistically significant**.
- But you then want to ask yourself whether that variable is **practically significant**: did R^2 increase (and the residual SE decrease) by enough to make it worth the more complex model?
- If we want to interpret the model, should we adjust for this variable *on substantive grounds*?
- **There is no universal rule (using p -values or otherwise) for adding or removing variables.**

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Adding new variables can change the existing coefficients

If we predict the banker's offer on just the max remaining case, the coefficient of maxleft is positive:

$$\widehat{\text{offer}} = 39173.27 + 0.05(\text{maxleft})$$



Adding new variables can change the existing coefficients

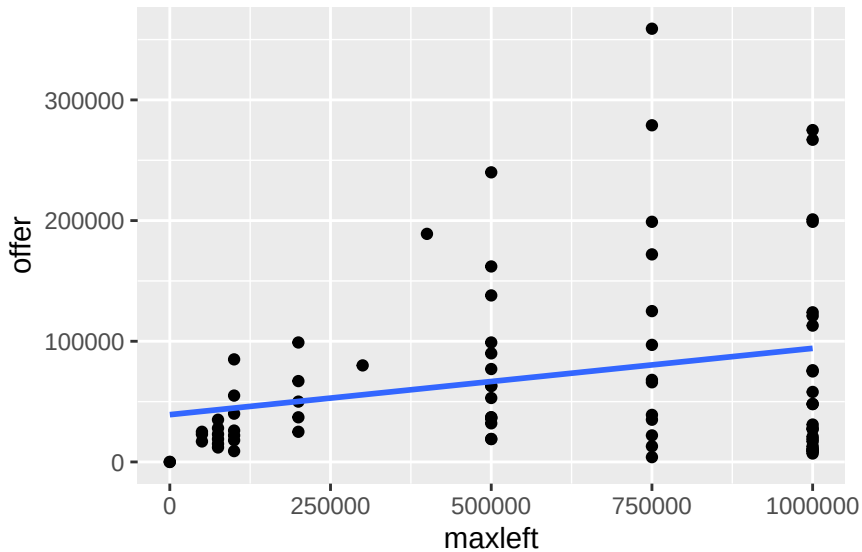
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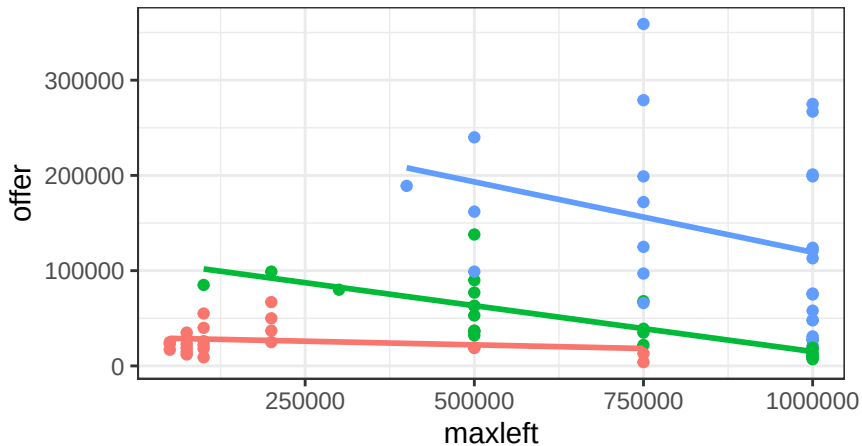
$$\widehat{\text{offer}} = 39173.27 + 0.05(\text{maxleft})$$

But if we add the average of the remaining cases, the coefficient of maxleft becomes negative:

$$\widehat{\text{offer}} = 6209.13 + 1.07(\text{average}) - 0.14(\text{maxleft})$$

What is going on here?





average_bucket Bottom 1/3 Mid 1/3 Top 1/3



What's going on?

- maxleft and average are correlated!
- At a given average, the dealer's offer will be lower if there is one large case and many small ones vs. several medium cases
- If average isn't in the model, maxleft does double duty – capturing the effect of maxleft, and the omitted variable average
- The two coefficients represent different things: In the simple model, the overall association between maxleft and offer, ignoring the average. In the multiple regression model, the association between maxleft and offer at a fixed value for the average.